

Multisensory Cybersecurity Authentication System: Blink-based Morse Encoding, EAR Liveness Detection and SHA-256 Verification

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Abstract—This paper presents a multisensory cybersecurity authentication system that combines visual blink patterns, liveness verification via the Eye Aspect Ratio (EAR), and cryptographic protection using SHA-256. The system converts intentional blink durations into Morse-like binary tokens (4-bit PINs), applies EAR-based temporal analysis for liveness detection, and secures stored credentials using SHA-256 hashing. We implemented a prototype using MediaPipe FaceMesh and OpenCV and evaluated it on a dataset of 20 participants. Under controlled conditions, the system achieved a detection accuracy of 95.6%, FAR of 2.1%, FRR of 4.3% and an average authentication time below 5 seconds. The approach addresses usability, accessibility and spoof-resistance concerns and demonstrates the feasibility of hands-free, sensory-driven authentication. (Detailed project report used as source). :contentReference[oaicite:2]index=2

Index Terms—Blink authentication, Eye Aspect Ratio (EAR), MediaPipe FaceMesh, liveness detection, SHA-256, multimodal biometrics

I. INTRODUCTION

Authentication mechanisms are under increasing pressure from attacks such as phishing, shoulder surfing, replay, and presentation attacks. While passwords and PINs are simple, they lack liveness guarantees and are often reused or weak. Physiological biometrics (fingerprint, face) improve convenience but can be spoofed with photos, masks, or replayed audio/video. Behavioral biometrics add an additional layer by exploiting user-specific patterns (gait, keystroke dynamics, gaze), but real-world systems must balance security with usability and accessibility.

This work introduces a *multisensory* approach emphasizing blink-based authentication as a primary modality. We encode blink durations using a Morse-like timing scheme (short/long $\rightarrow$ 0/1), verify liveness through EAR time-series analysis, and protect templates using cryptographic hashing (SHA-256). The approach is particularly suited to hands-free or accessibility-aware settings, and resists simple spoofing attempts due to temporal dynamics requirements. The extended theory, experimental design, and evaluation are drawn and expanded from the project report. :contentReference[oaicite:3]index=3

II. RELATED WORK

A. Password and Knowledge-based Methods

Studies show that passwords suffer from reuse and predictability [1], [2]. These weaknesses motivate biometric and behavioral replacements or augmentations.

B. Biometrics and Multimodal Fusion

Physiological biometrics offer unique user traits but can be vulnerable to presentation attacks (printed photos, masks). Multimodal systems — combining face, voice, gait — reduce false acceptance and increase resilience [4]. Recent multisensory systems fuse visual, audio and behavioral signals to defend against deepfakes and video injection [?], [9].

C. Eye-based and Behavioral Authentication

The Eye Aspect Ratio (EAR) method formalized by Soukupová and Čech [5] provides a computationally efficient blink detection mechanism using facial landmarks. Zhang et al. (2020) demonstrated blink-to-PIN authentication combining EAR with vision pipelines [6]. Behavioral biometrics such as blink timing provide cognitive-behavioral signals that are difficult to reproduce without real-time control.

D. Liveness Detection and Anti-Spoofing

Liveness detection strategies use motion, micro-expressions or challenge-response behaviors to distinguish live users from spoof artifacts [?], [?]. Temporal analysis of blink dynamics is a robust liveness cue when combined with facial movement analysis.

E. Cryptography for Template Protection

Hashing (SHA-family) and keyed-hashing (HMAC) are standard for irreversible template storage. Using SHA-256 to protect blink PINs ensures that stored templates cannot be easily reversed if the database is compromised [8].

III. SYSTEM ARCHITECTURE AND METHODOLOGY

The architecture is modular: Capture $\rightarrow$ Preprocess $\rightarrow$ Landmark Extraction $\rightarrow$ EAR computation $\rightarrow$ Blink segmentation and encoding $\rightarrow$ Hashing and verification. Figure placeholders (from the project PDF) illustrate registration, live detection and console outputs (Figures 1–3).

A. Signal Acquisition

Video frames are acquired at 30 FPS. Each frame is normalized to 640×480 for consistent processing. Optional microphone input is reserved for future fusion but was not used in the present experiments.

B. Facial Landmark Extraction

MediaPipe FaceMesh provides dense facial landmarks (468 points) in real time. We select eye-related landmark indices to compute EAR for both eyes and average them to obtain a robust openness measure.

C. EAR Computation and Thresholding

For each eye, the EAR is computed:

$$EAR = \frac{\|p_2 - p_6\| + \|p_3 - p_5\|}{2\|p_1 - p_4\|}$$

We choose a base threshold $T_{EAR} = 0.25$ to indicate eye closure; this is dynamically calibrated per user by measuring baseline open-eye EAR during registration. Temporal smoothing (5-frame moving average) reduces false positives caused by micro-movements.

D. Blink Segmentation and Encoding

A blink is defined as an event where the smoothed EAR falls below T_{EAR} and then returns above it. The blink duration is measured from EAR drop to EAR recovery. Classification rule:

- Blink duration < 0.4 s $\Rightarrow$ bit ‘0’ (quick)
- Blink duration ≥ 0.4 s $\Rightarrow$ bit ‘1’ (long)

Four intentional blinks generate a 4-bit PIN token (e.g., 0100).

E. Hashing and Verification

During enrollment the 4-bit PIN is concatenated as a string and hashed using SHA-256 (Python’s hashlib) and stored in `users.json`. During authentication the newly generated PIN is hashed and compared. If equal, access is granted.

F. Pseudocode: Blink-based Authentication

- 1: **Enroll**(username):
- 2: Capture baseline EAR for calibration
- 3: Prompt user to perform 4 intentional blinks
- 4: For each blink, measure duration d_i
- 5: Map durations to bits: $b_i \leftarrow (d_i \geq 0.4)$
- 6: PIN $\leftarrow \text{concat}(b_1, b_2, b_3, b_4)$
- 7: Store SHA-256(PIN) in `users.json`
- 8: **Authenticate**(username):
- 9: Retrieve stored hash
- 10: Capture blinks and compute PIN as above
- 11: If SHA-256(PIN) == stored hash then grant access

IV. DATASET GENERATION

Because public multisensory datasets with blink-Morse encoding are rare, we collected a custom dataset as follows (see project report for raw logs) [10]:

- 20 participants
- 5 trials per lighting condition (bright, low-light, side-lit)
- Webcam 30 FPS, resolution 640×480
- For each trial we recorded raw frames, EAR series, blink timestamps and the resulting 4-bit PIN

This dataset enabled per-user threshold calibration and measurement of EAR stability across conditions.

V. PREPROCESSING AND FEATURE EXTRACTION

Preprocessing included resizing, color conversion and temporal smoothing. Features extracted:

- EAR time-series (smoothed)
- Blink onset and offset times
- Blink durations and inter-blink intervals
- Encoded 4-bit PIN

We also compute an EAR Stability Index (variance of EAR during open-eye periods) to measure robustness to noise.

VI. IMPLEMENTATION DETAILS

The prototype was implemented in Python 3.11 using:

- OpenCV 4.x for capture and GUI
- MediaPipe 0.10 for FaceMesh
- NumPy for numerical ops
- hashlib for SHA-256 hashing
- JSON for lightweight credential storage

The GUI overlays EAR, threshold, and PIN progress to provide immediate user feedback.

VII. EXPERIMENTS AND EVALUATION

A. Setup

Hardware: Intel Core i7 (12th Gen), 16 GB RAM, 1080p webcam. Lighting ranges: 500–800 lux for controlled conditions.

B. Metrics

We report:

- Detection Accuracy = $\frac{\text{correct detections}}{\text{total events}}$
- False Acceptance Rate (FAR)
- False Rejection Rate (FRR)
- Average authentication time
- EAR Stability Index

C. Results

TABLE I: Average performance across participants (20 users)

Metric	Value	Remarks
Detection Accuracy	95.6%	High in stable lighting
False Acceptance Rate (FAR)	2.1%	Low risk
False Rejection Rate (FRR)	4.3%	Occasional due to timing
Avg. Authentication Time	4.8 s	Real-time friendly
EAR Stability Index	± 0.03	Controlled illumination

Detailed observations:

- Glasses/reflections increase false rejections by altering EAR measurements.
- Users required 3 trials to adapt to the timing constraints; average time decreased with practice.
- The system resists static-photo spoofing due to temporal dynamics; video-replay attacks require synchronized blink timing and are harder to execute without physical control.

VIII. COMPARATIVE ANALYSIS

Compared against simple password and face-only systems:

- Password/PIN: deterministic but lacks liveness and is vulnerable to theft.
- Face-only biometric: $\sim 92\%$ accuracy in our comparisons but vulnerable to photo spoofing.
- Proposed method: 95.6% accuracy plus strong liveness verification due to EAR temporal constraints.

IX. USER INTERFACE AND FIGURES

The following figures are cropped from the project PDF and embedded directly. They show the registration console (EAR readings), live detection overlay, and authentication console output.

```
[DEBUG] Current EAR: 0.306 (Threshold: 0.25)
[BLINK START] EAR: 0.140
[DETECTED] QUICK blink (0.24s) -> 0
[DEBUG] Current EAR: 0.256 (Threshold: 0.25)
[BLINK START] EAR: 0.133
[DEBUG] Current EAR: 0.116 (Threshold: 0.25)
[DETECTED] LONG blink (0.80s) -> 1
[BLINK START] EAR: 0.171
[DETECTED] QUICK blink (0.10s) -> 0
[BLINK START] EAR: 0.187
[DETECTED] QUICK blink (0.06s) -> 0
[INFO] PIN capture complete!

Blink sequence: ['quick', 'long', 'quick', 'quick']
PIN: 0100
[SUCCESS] PIN registered for user: test username
```

Fig. 1: Registration console: EAR values, blink durations and classification (quick/long).



Fig. 2: Real-time blink detection overlay showing EAR, threshold and PIN progress.

```
[DEBUG] Current EAR: 0.267 (Threshold: 0.25)
[BLINK START] EAR: 0.244
[DEBUG] Current EAR: 0.175 (Threshold: 0.25)
[DETECTED] LONG blink (0.89s) -> 1
[INFO] Resetting sequence...
[DEBUG] Current EAR: 0.300 (Threshold: 0.25)
[BLINK START] EAR: 0.154
[DETECTED] QUICK blink (0.26s) -> 0
[BLINK START] EAR: 0.157
[DEBUG] Current EAR: 0.083 (Threshold: 0.25)
[DETECTED] LONG blink (1.23s) -> 1
[DEBUG] Current EAR: 0.302 (Threshold: 0.25)
[DEBUG] Current EAR: 0.308 (Threshold: 0.25)
[BLINK START] EAR: 0.163
[DETECTED] QUICK blink (0.12s) -> 0
[DEBUG] Current EAR: 0.320 (Threshold: 0.25)
[BLINK START] EAR: 0.118
[DETECTED] QUICK blink (0.21s) -> 0
[INFO] PIN entry complete!

Blink sequence: ['quick', 'long', 'quick', 'quick']
Entered PIN: 0100

=====
🎉 AUTHENTICATION SUCCESS! 🎉
Welcome back, test_username!
=====

Session Summary:
- Blinks detected: 4
- EAR threshold: 0.25
- Duration threshold: 0.4s
```

Fig. 3: Authentication console output demonstrating successful PIN hash match.

X. SECURITY AND PRIVACY CONSIDERATIONS

- **Template protection:** SHA-256 is used to avoid storing raw PINs. For extra safety use salted hashing or HMAC.
- **Anti-replay:** Temporal EAR dynamics and challenge-response blinking increase difficulty of replay attacks. Future work should incorporate micro-expression checks and head-motion correlation.
- **Storage:** Prototype uses local JSON. Production use requires encrypted databases, access controls and secure channels (TLS).
- **Accessibility:** The hands-free nature is beneficial for users with motor impairments, but alternative paths (voice, hardware token) should be offered when blinking is impractical.

XI. LIMITATIONS

- Performance degrades with occlusion (glasses glare), extreme head rotation and very low light.

- Fixed timing thresholds may not generalize — adaptive per-user calibration is recommended.
- Dataset size (20 users) is modest; larger datasets improve generalization and allow ML-based classifiers.

XII. CONCLUSION AND FUTURE WORK

We presented a multisensory blink-based authentication system combining EAR liveness detection and SHA-256 template protection. Results indicate practical accuracy and low latency, making it suitable for accessibility-driven authentication. Future work includes:

- Adaptive EAR calibration (per-user baselining)
- CNN/LSTM-based blink classifiers to replace static thresholds
- Multimodal fusion with voice/EEG/gesture
- Edge deployment and secure cloud synchronization
- Open-source dataset release for benchmarking

ACKNOWLEDGMENT

The authors thank Dr. P. Gangamohan for guidance and all participants for contributions to the dataset. The PBL project report served as the primary source for this paper.

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